

Unit II - Class Activity 1

Question 1: Hospital Patient Admission Analysis

R Program:

```
library(ggplot2)

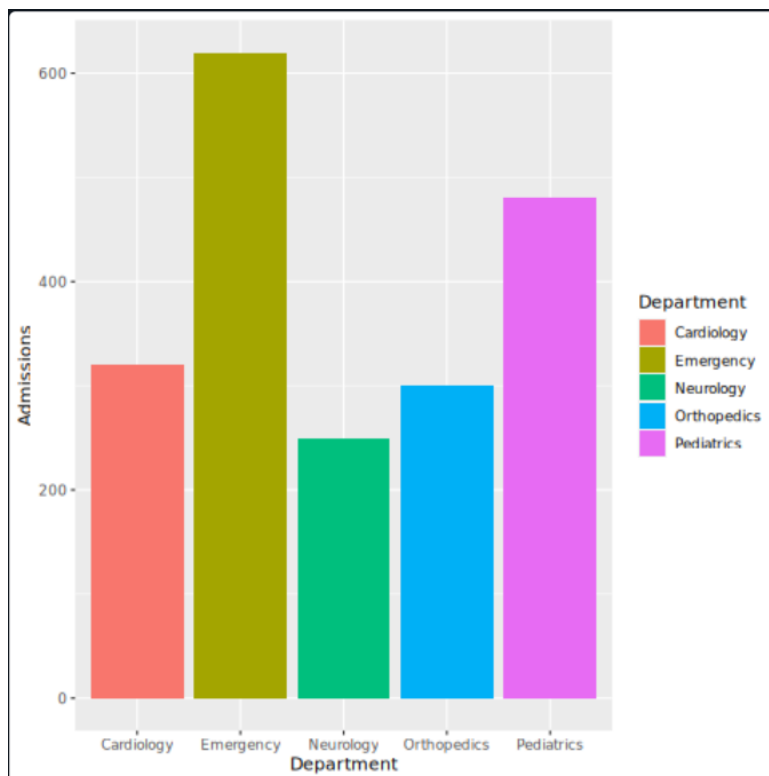
Department <- c("Cardiology","Neurology","Pediatrics",
               "Orthopedics","Emergency")
Admissions <- c(320,250,480,300,620)

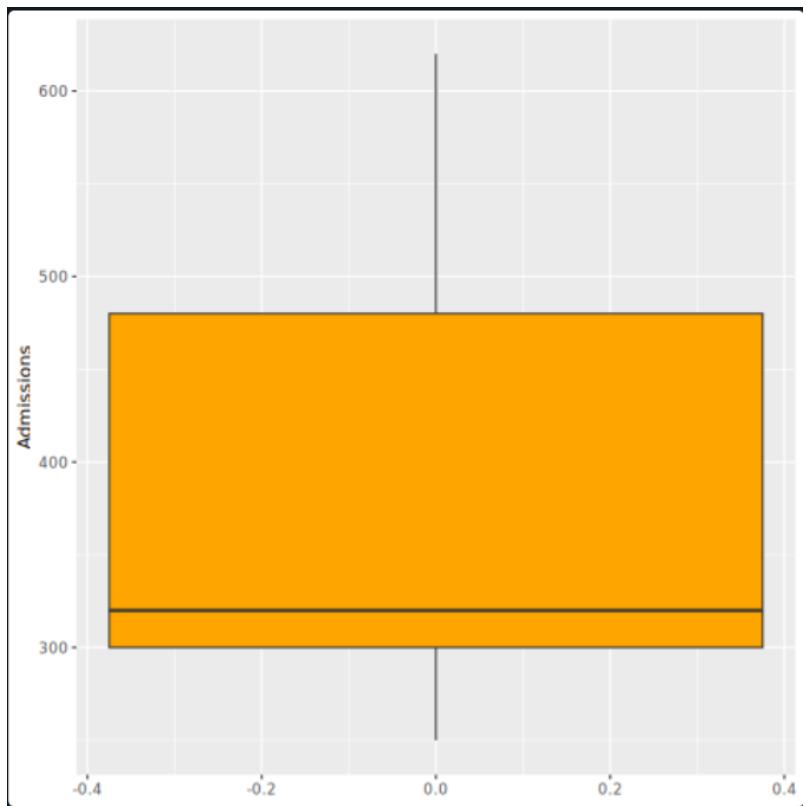
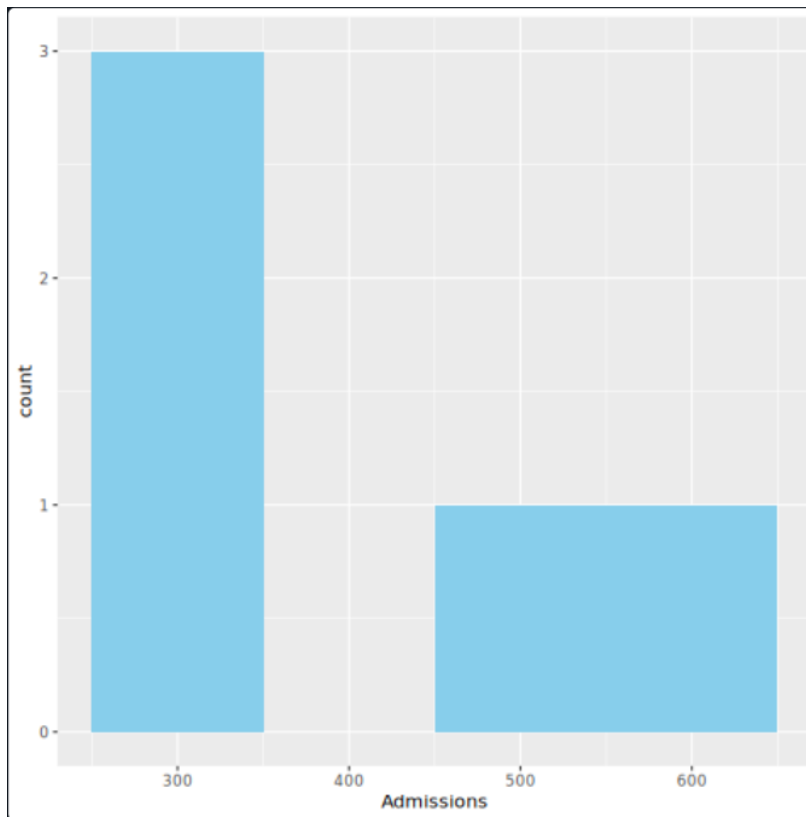
df <- data.frame(Department,Admissions)

ggplot(df,aes(Department,Admissions,fill=Department))+
  geom_bar(stat="identity")

ggplot(df,aes(Admissions))+
  geom_histogram(binwidth=100,fill="skyblue")

ggplot(df,aes(y=Admissions))+
  geom_boxplot(fill="orange")
```





Question 2: E-Commerce Customer Spending Behaviour

R Program:

```
library(ggplot2)

Customer=c("A","B","C","D","E")
Spending=c(1200,800,1500,500,2200)
Orders=c(15,9,20,6,28)
Category=c("Electronics","Fashion","Electronics","Grocery","Fashion")

df=data.frame(Customer,Spending,Orders,Category)

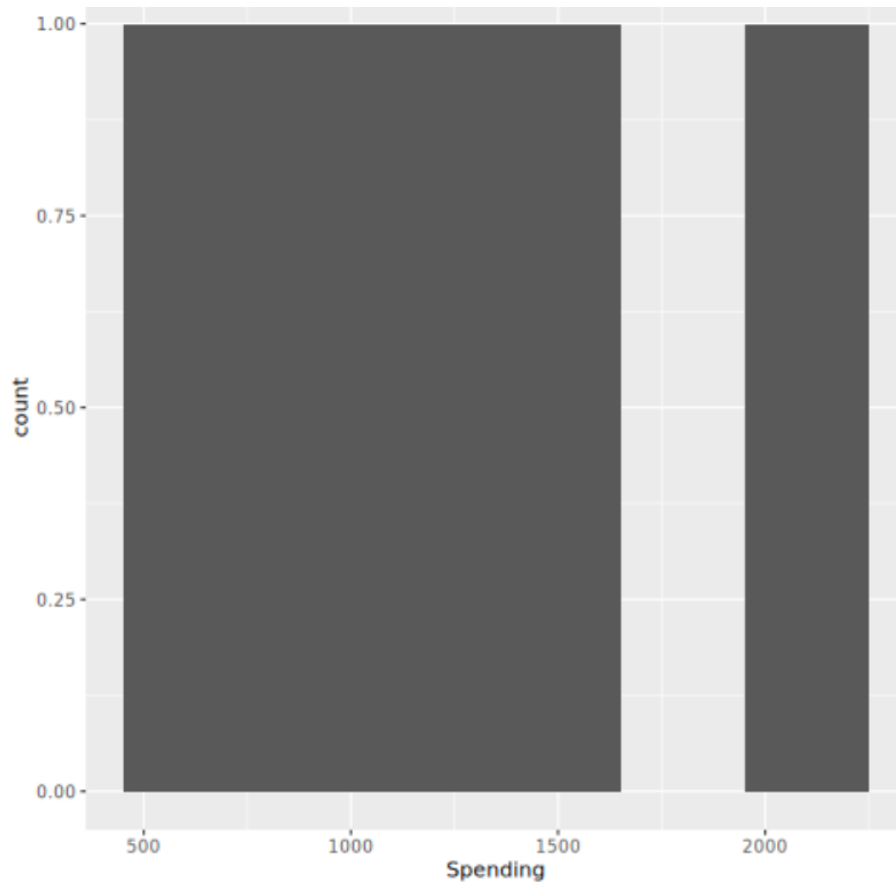
ggplot(df,aes(Spending))+geom_histogram(binwidth=300)

ggplot(df,aes(y=Spending))+geom_violin(fill="pink")

ggplot(df,aes(y=Spending))+geom_boxplot()

ggplot(df,aes(Spending))+geom_density(fill="lightblue")

ggplot(df,aes(Category,Spending,fill=Category))+geom_bar(stat="identity")
```



Question 3: University Examination Performance Dashboard

R Program:

```
library(ggplot2)
```

```
Department=c("CSE","IT","ECE","EEE","AIML")
```

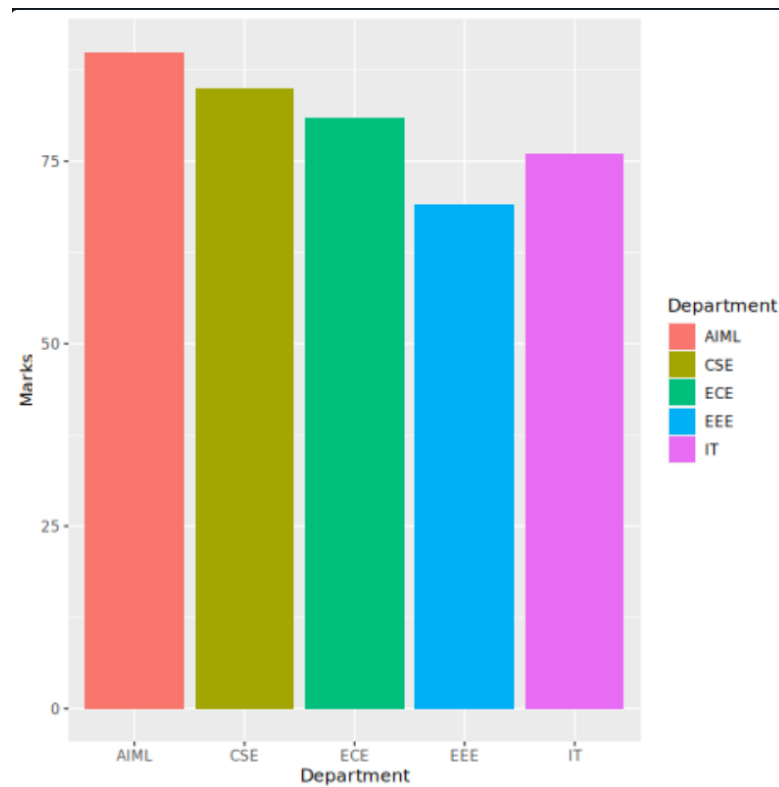
```
Marks=c(85,76,81,69,90)
```

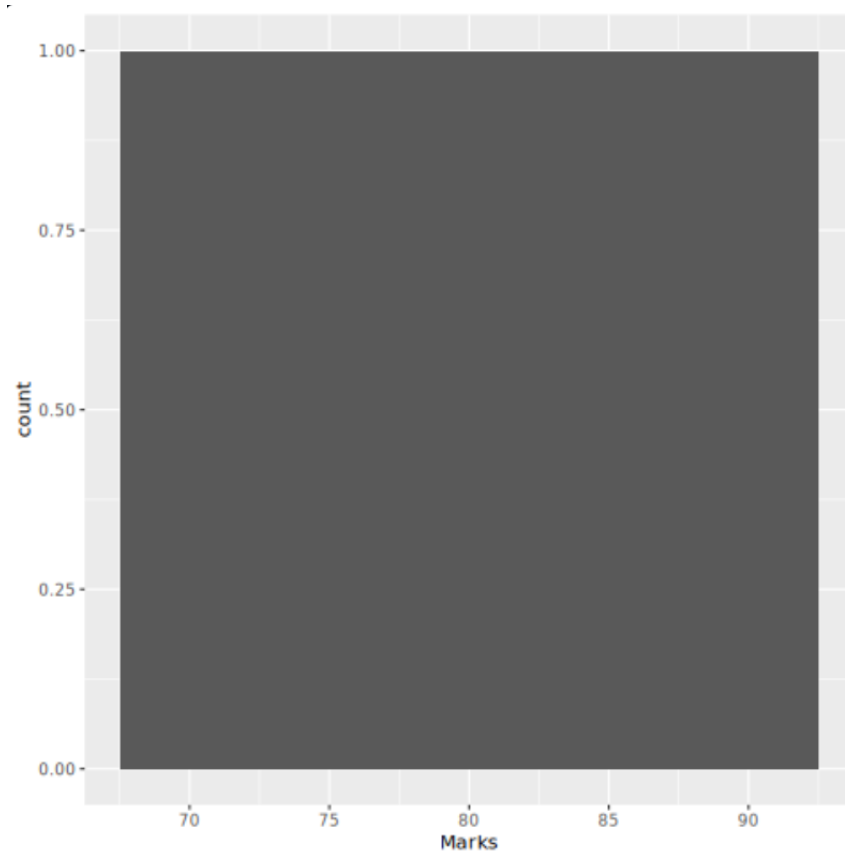
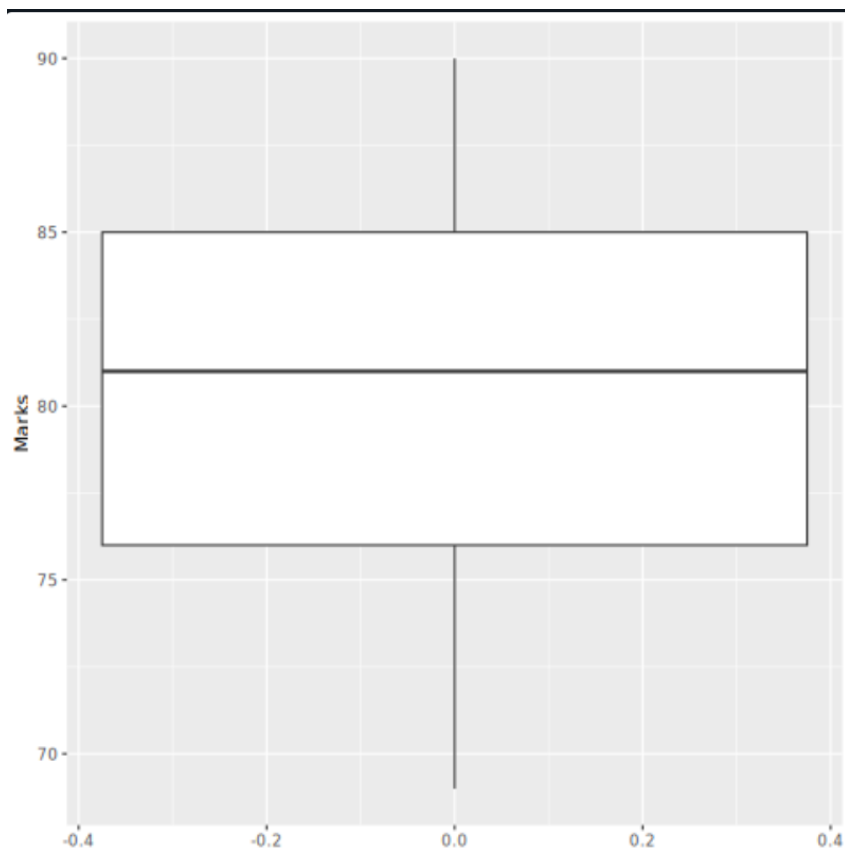
```
df=data.frame(Department,Marks)
```

```
ggplot(df,aes(Department,Marks,fill=Department))+geom_bar(stat="identity")
```

```
ggplot(df,aes(y=Marks))+geom_boxplot()
```

```
ggplot(df,aes(Marks))+geom_histogram(binwidth=5)
```





Question 4: Climate and Rainfall Distribution Study

R Program:

```
library(ggplot2)

City=c("Chennai","Delhi","Mumbai","Kolkata","Bangalore")
Rainfall=c(120,80,210,150,95)
Temperature=c(33,36,31,32,28)
Humidity=c(75,60,85,80,65)

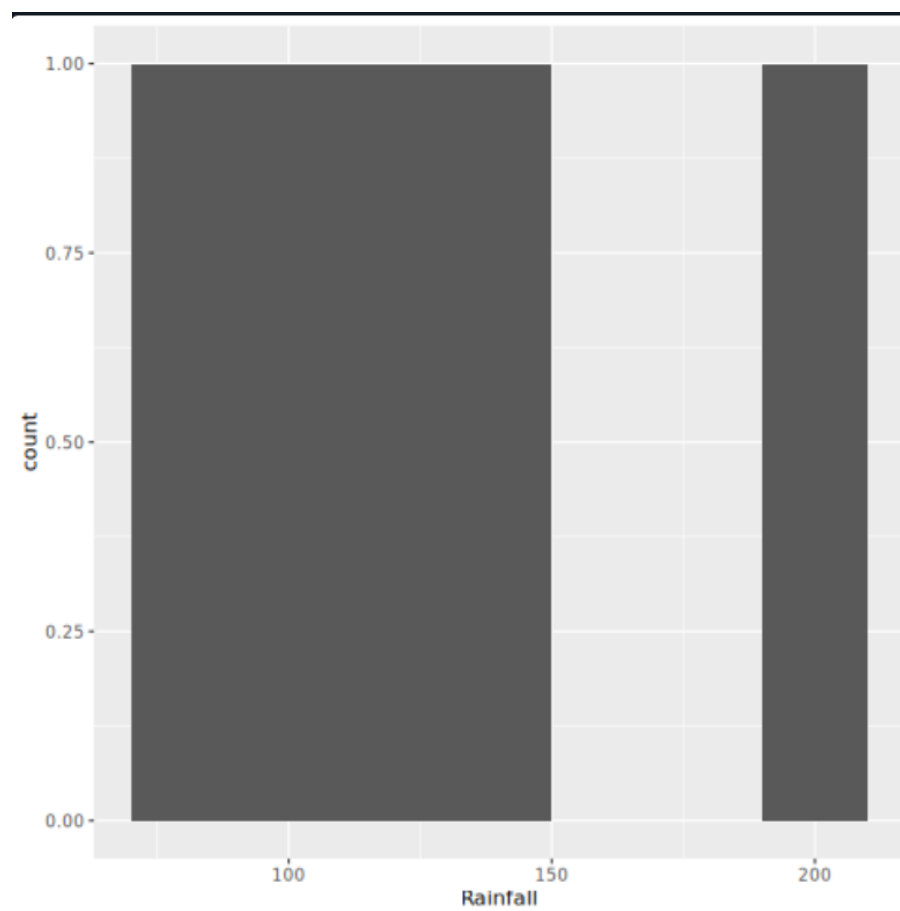
df=data.frame(City,Rainfall,Temperature,Humidity)

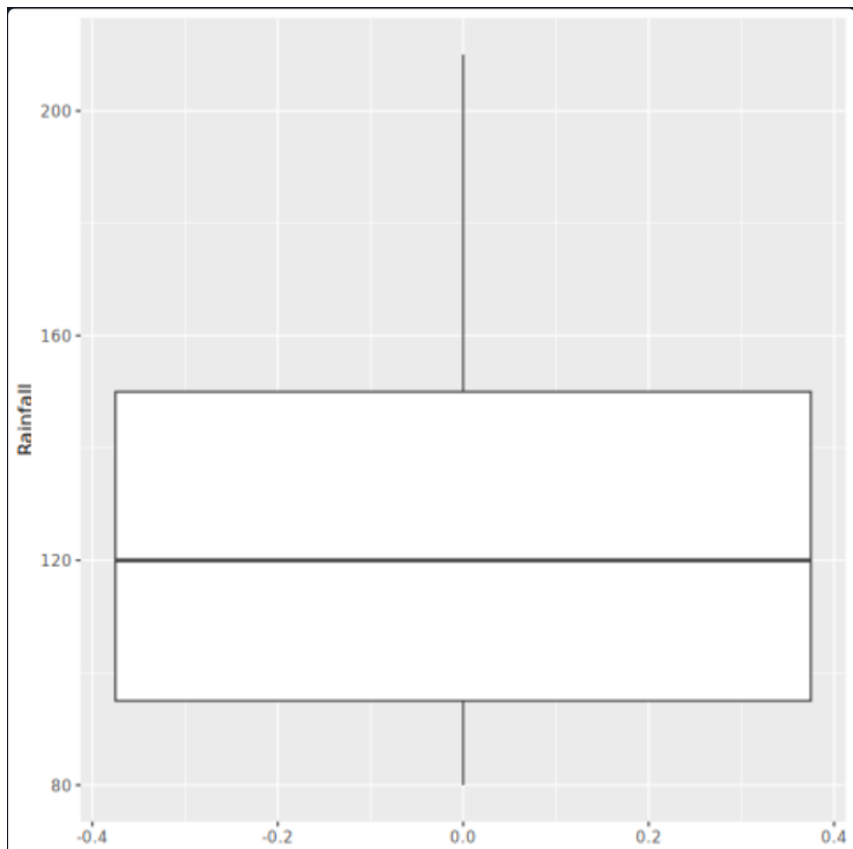
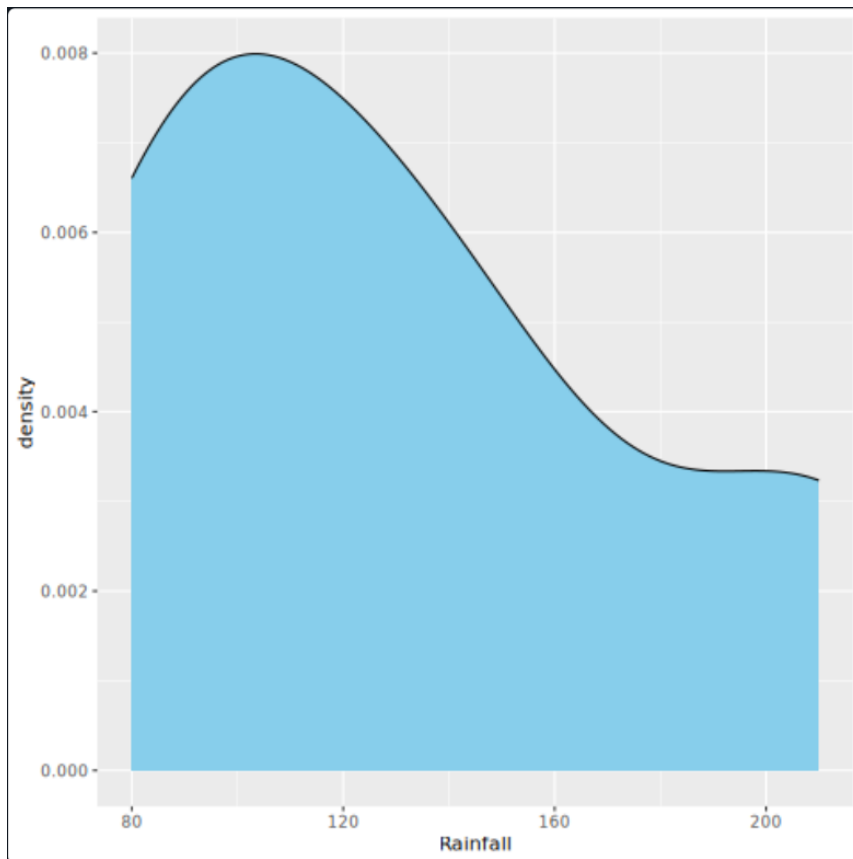
ggplot(df,aes(Rainfall))+geom_histogram(binwidth=20)

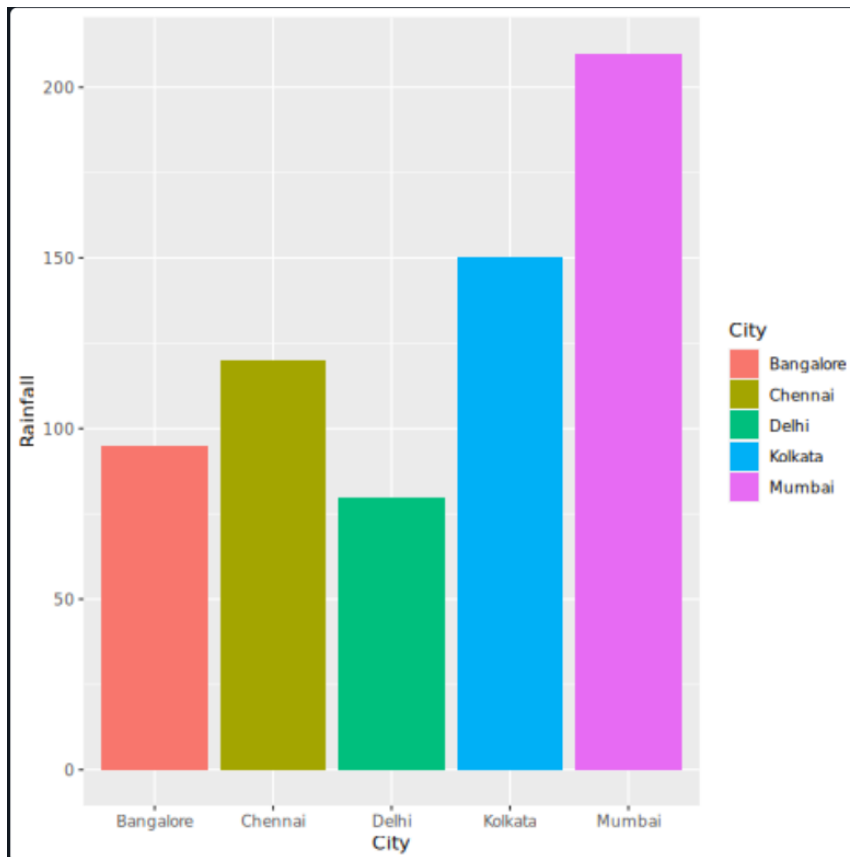
ggplot(df,aes(Rainfall))+geom_density(fill="skyblue")

ggplot(df,aes(y=Rainfall))+geom_boxplot()

ggplot(df,aes(City,Rainfall,fill=City))+geom_bar(stat="identity")
```







Question 5: Bank Loan Risk Assessment

R Program:

```
library(ggplot2)
```

```
Customer=c("A","B","C","D","E")
```

```
Loan=c(500000,200000,900000,150000,650000)
```

```
Income=c(800000,350000,1200000,280000,900000)
```

```
CreditScore=c(780,620,810,590,760)
```

```
df=data.frame(Customer,Loan,Income,CreditScore)
```

```
ggplot(df,aes(Loan))+geom_histogram(binwidth=100000)
```

```
ggplot(df,aes(y=Loan))+geom_boxplot()
```

```
ggplot(df,aes(Income,Loan,color=CreditScore))+geom_point(size=4)
```

